



Longitudinal comparison of low- and high-velocity resistance training in relation to body composition and functional fitness of older adults

Michelle Gray¹ · Melissa Powers² · Larissa Boyd² · Kayla Garver³

Received: 3 October 2017 / Accepted: 6 March 2018 / Published online: 22 March 2018
© Springer International Publishing AG, part of Springer Nature 2018

Abstract

Background Functional mobility disability affects more than one in five adults over 70 years and increases to 80% by 90 years. While negative changes in mobility are multifactorial, deleterious body composition changes contribute significantly. Resistance training alters the negative trajectory of physical function as well as increases lean mass among older adults. Recently, high-velocity (HV) resistance training has been indicated as an effective intervention to increase lean mass and functional performance.

Aims The present investigation compared body composition, physical function, and muscular strength changes between HV and LV resistance training programs.

Methods Participants > 65 years ($n = 53$) were randomly assigned to LV, HV, or active control (AC) group and participated in their respective intervention for 48 weeks.

Results Analysis of covariance revealed no significant body composition changes over time between groups ($p > 0.05$). Eight-foot up-and-go performance improved in the HV and AC groups ($p < 0.05$) with no change in the LV group ($p > 0.05$) over time. Muscular strength increased in both the LV and HV groups within the first 24 weeks, while only in the LV group, muscular strength continued to increase from 24 to 48 weeks ($p < 0.05$).

Discussion Resistance training appears to be an effective intervention for improving aspects of physical function and muscular strength; however, no significant changes in body composition were observed over the 48-week intervention.

Conclusion Findings from the current investigation support use of resistance training for improving physical function among community-dwelling older adults.

Keywords High-velocity resistance training · Physical function · Body composition · Muscular strength

Introduction

More than 20% of older adults (≥ 65 years) reporting difficulty performing at least one aspect of daily living [1]. The negative impact of mobility limitations are vast leading to reduced quality of life [2], increased entry into

long-term care facilities, and greater risk of falls [3–5]. While the mechanisms by which these mobility limitations occur are multifactorial and unclear, losses in lean mass (sarcopenia) coupled with increases in fat mass (obesity) have been identified as significant contributors [6]. Thus, interventions known to combat these deleterious changes are imperative.

While age is charged as a primary factor in lean mass loss, more recent evidence suggests reductions in physical activity may better predict loss when compared to age alone [7–10]. With age, there is a reduction in all types of physical activity participation, with the greatest reduction occurring among muscle strengthening activities. In fact, only 23% of adults 65 years or older self-report regularly participating in muscle strengthening exercises [11]. Reasons for the lack of exercise participation are vast and include perceived barriers [12] and difficulty of the task being performed. Older

✉ Michelle Gray
rgray@uark.edu

¹ Exercise Science Research Center, Office for Studies on Aging, University of Arkansas, HPER 321, Fayetteville, AR 72701, USA

² Kinesiology and Health Studies, University of Central Oklahoma, 100 N. University Avenue, Edmond, OK 73034, USA

³ University of Oklahoma Medical Center, Oklahoma City, OK, USA

adults that participate in physical activity often choose aerobic activities such as walking, because it is both practical and beneficial [13]. Additionally, resistance-type activities are prescribed in such a way that are perceptually more difficult to perform when compared to walking or other aerobic activities [14]. Decreasing the intensity of the resistance training intervention could promote participation as well as increase functional outcomes.

Strength exercises can be divided into low- and high-velocity movements; low-velocity resistance training is performed at a rate of 2 s per lifting phase (concentric and eccentric), while high-velocity resistance training requires the participant to move as quickly as possible during the concentric phase and slowly (2 s) for the eccentric phase of the lift. High-velocity resistance training produces the greatest increases in functional outcomes among older adults [15, 16] when compared to low-velocity movements. While high-velocity resistance training can be performed on the same equipment as low-velocity resistance training, most studies among older adults use pneumatic resistance equipment which is expensive and not widely available in community settings. One research group studied a community-based high-velocity resistance training program utilizing weighted vests (InVEST) [17]. InVEST was compared to a resistance exercise program performed at low velocities, but at a similar exercise intensity. Both groups improved similarly on the functional outcomes, but effect sizes indicated a trend toward greater functional improvements in the InVEST group. The benefit of this study was the use of free weight resistance (i.e., weighted vests) which is more readily available in the community than traditional high-velocity resistance training machines (i.e., pneumatic equipment).

Research indicates resistance training as one of the more effective and safe intervention for off-setting the deleterious effects of declines in lean, fat, and bone mass [18]. For decades, progressive low-velocity resistance training, especially at higher intensities (80% of maximum), has been recommended for older adults to improve body composition as well as functional outcomes [19–25] such as stair climbing, walking speed, and chair rise performance [19, 23, 24]. However, among older adults, the effects of resistance training on lean and fat mass are equivocal; although results primarily report reduced fat mass and increased lean mass [19, 24, 26–28]. Recent evidence suggests that high-velocity resistance training results in greater benefits than traditional low-velocity resistance training for increasing lean mass, but results remain inconsistent [29, 30].

The ideal exercise intensity for older adults performing high-velocity resistance training remains unknown. While exercise intensity recommendations for older adults during low-velocity resistance training are 60–80%

of maximum (i.e., high-intensity), the recommended intensity during high-velocity training is much lower at 40–60% of maximum [31]. These recommendations are based on little evidence and are mostly due to concerns for the safety of older participants. In 2005, de Vos and colleagues found that muscle power improvements were not dependent upon exercise intensity (20 versus 80% maximum), but strength and endurance gains were much greater at higher intensities [32]. More recently, the tolerance of high-velocity movements among healthy adults over the age of 80 years has been examined and found beneficial for increasing muscular power [33]. This type of training produced no negative neuromuscular effects, but caution against generalizations across all elder groups and settings is warranted.

With the increasing popularity of high-velocity resistance training for older adults [14, 15, 32, 33], it is imperative that this type of training be studied in a community-based setting using a resistance source that is widely available to older adults. Therefore, the primary purpose of this study was to compare high-velocity resistance training to low-velocity resistance training to determine the impact on body composition and physical function in adults over age 65 from community-based settings.

Method

Procedures

This study was approved by the Institutional Review Board (IRB # 08149) at a midwest regional institution. Before any assessments were completed, all subjects were informed of their rights as a research subject and signed the approved informed consent form and received a medical release by their primary care physician. After signing the informed consent document, the physician clearance form was faxed to the subject's primary care physician and returned to our lab. After receiving physician clearance, subjects were randomly assigned to one of three training groups: traditional resistance training at low-velocity (LV), high-velocity resistance training (HV), or an active control group (AC) using a random numbers generator.

Participants

Participants were male and female volunteers with a minimum age of 70 years. Inclusion criteria for all participants included: (1) normal cognitive function (Mini Mental State Exam > 23) [34]; (2) no diagnosis of unstable or unmanaged cardiovascular disease, hypertension, or diabetes; (3) had not participated in resistance or

strengthening exercise within the previous 12 months; (4) no current diagnosis of neuromuscular or musculoskeletal disease or injury prohibiting exercise participation; and (5) received medical clearance from the subject's primary care physician.

Assessments

Body composition

A total-body dual-energy X-ray absorptiometry (iDXA; GE Healthcare, Buckinghamshire, UK) scan was conducted to assess measures of total-body lean-tissue mass, total-body fat mass, total-body mass, total-body bone mineral content, and percent body fat (BF%). The iDXA utilized underwent full calibration procedures every 3 months and quality assurance analysis each morning before any measurements. All assessments were conducted at baseline, 24, and 48 weeks.

Physical function

Physical function was determined via the senior fitness test (SFT) battery [35]. SFT includes assessments of upper- and lower-body strength (arm curl and chair stand, respectively), agility (eight-foot up-and-go), cardiorespiratory endurance (6-min walk test), and upper- and lower-body flexibility (back scratch and chair sit-and-reach, respectively). SFT has previously established test–retest reliability (0.93–0.97) for all measures [36].

1 repetition maximum

One repetition maximum (1 RM) was evaluated according to published standards [37] for the following exercises: chest press, row, biceps curl, triceps extension, lateral raise, calf raise, and knee curl. All exercises were performed with free weights to mimic the exercise modality used during training. One RM testing was re-assessed every 6 weeks to maintain proper exercise intensity throughout the study's duration.

Hand-grip

Grip strength was assessed using a digital hand-grip dynamometer (Takei Scientific Instruments; Niigata, Japan). The dynamometer was fit to each participant's hand, so that their middle joint of the middle phalanx was at a 90° angle. Each participant was instructed to squeeze maximally for at least 3 s, while a researcher provided standard encouragement. Three trials were performed alternating left and right hands with a 60-s rest between trials. Hand-grip dynamometry has been determined to be a valid and reliable measure

of upper-body strength [38] and measure of physical function among older adults [39, 40].

Muscular power

Lower-body muscular power was measured during a sit-to-stand task. Each participant was seated in a standard chair (0.43 m seat height) with both feet placed flat on the floor. With their arms crossed over their chest, they were instructed to stand as quickly and as safely as possible. Lower-body muscular power was measured using a Tendo Weightlifting Analyzer (TENDO Sports Machines, Slovak Republic). Participants completed one set of ten sit-to-stand tasks with a 60-s rest between trials. Average of the ten trials was used in analyses. This assessment has been found to be a valid assessment of lower-body muscular power and test–retest reliability is high (Cronbach's $\alpha = 0.98$) [41].

Exercise intervention

The exercise intervention was conducted in a community setting, using free weight resistance for 48 weeks. Each group participated in group exercise 2 days per week for approximately 60 min per session. For the LV and HV groups, each training session began with a 5-min warm-up and ended with a 5-min stretching routine. Each exercise session was supervised by a member of the research team for increased safety as well as to document attendance and compliance.

The LV group performed each concentric and eccentric movement at a rate of 2 s per phase at 80% of their 1RM. To maintain training intensity, 1RM was re-assessed every 6 weeks. For safety, the HV group trained at a low-velocity at 80% 1RM for 24 weeks before transitioning to high-velocity training. This prolonged familiarization was designed to adequately increase muscular strength before prescribing plyometric-type activities for this population. Following familiarization, the HV group trained at 50% of 1RM for the remaining 24 weeks. After the familiarization period, the HV group was instructed to lift the weight concentrically “as quickly as possible”, followed by a 2-s eccentric phase. The active control group (AC) performed all exercises without any external resistance at a rate of 2 s for both the concentric and eccentric phases. The eight exercises performed by all groups were: seated chest press, seated bent-over row, biceps curl, overhead triceps extension, lateral raise, standing knee curl, heel raises, and chair stand or half lunge. The overhead triceps extension and lateral raise were modified for the HV group. The triceps extension was performed by standing with a medicine ball at chest level. Participants explosively threw the ball to the ground by extending the elbows. The lateral raise was

performed by quickly tossing a single-handled medicine ball at the end of the lateral raise range of motion. Each supervised training session last approximately 60 min and each group completed three sets of ten repetitions at least 2 days per week for 48 weeks.

Statistical analysis

All statistical procedures were performed using SPSS (v.23, IBM, Armonk, NY). Normal distribution was determined using histograms and descriptive statistics. Based on a priori calculations, to have adequate statistical power of 80%, total needed sample size is 93. Pre-test data were analyzed using a one-way ANOVA to determine differences between groups at baseline (Table 1). Repeated measures ANCOVA with the baseline score as the covariate were used to determine

changes between groups over time (24 and 48 weeks). Statistical significance was set at $p < 0.05$; results are presented as means $\pm$ SD. Where warranted, post hoc analyses will be performed using the Bonferroni correction to protect against an inflated Type I error rate.

Results

Participants

Ninety-nine participants with a mean age of 81.2 ± 5.6 years completed all baseline testing. Participants were randomly assigned to one of three groups: HV ($n=41$), LV ($n=34$), or AC ($n=24$). The sample was predominately female (68%). Participant characteristics are located in Table 1.

Table 1 Baseline characteristics of the study population

Variable	Group			
	HV $n=34$	LV $n=41$	AC $n=24$	
Female (%)	70.6	75.6	70.8	
Age (years)	81.6 ± 5.9	81.0 ± 5.5	81.3 ± 5.3	
Height (cm)	169.99 ± 9.04	165.78 ± 9.30	163.77 ± 5.84	
Weight (kg)	76.41 ± 14.17	73.17 ± 13.27	68.21 ± 10.94	
Body fat (%)	40.54 ± 8.26	40.56 ± 7.64	38.53 ± 8.62	
Fat mass (kg)	66.25 ± 22.01	62.30 ± 16.76	55.69 ± 18.32	
Lean-tissue mass (kg)	94.90 ± 17.90	90.41 ± 1.26	86.58 ± 14.30	
Bone mineral content (kg)	5.36 ± 1.38	5.14 ± 1.24	4.83 ± 0.84	
Chair stand (# in 30 s)	13.0 ± 4.4	12.5 ± 3.5	11.1 ± 5.6	
Arm curl (# in 30 s)	18.3 ± 4.8	17.5 ± 3.3	15.9 ± 6.7	
UPGO (sec)	7.16 ± 3.00	6.82 ± 2.44	8.03 ± 4.59	
Variable	Group			<i>p</i> value
	HV $n=20$	LV $n=25$	AC $n=8$	
Female (%)	70.0	68.0	62.5	
Age (years)	82.1 ± 5.9	81.4 ± 6.0	80.0 ± 5.6	0.84
Height (cm)	172.82 ± 9.28	164.44 ± 8.73	164.29 ± 7.06	0.28
Weight (kg)	65.52 ± 7.60	72.12 ± 11.75	76.18 ± 12.23	0.13
Body mass index (kg/m ²)	27.22 ± 3.19	27.65 ± 3.83	24.33 ± 4.75	0.27
Body fat (%)	40.54 ± 8.26	40.75 ± 7.67	37.94 ± 8.33	0.06
Fat mass (kg)	63.96 ± 17.14	61.38 ± 16.72	45.72 ± 9.47	0.01
Lean-tissue mass (kg)	96.16 ± 20.76	90.35 ± 17.51	86.49 ± 9.20	0.49
Bone mineral content (kg)	5.77 ± 1.52	5.21 ± 1.31	4.88 ± 0.69	0.37
Chair stand (# in 30 s)	12.3 ± 5.3	12.8 ± 3.7	13.4 ± 3.6	0.95
Arm curl (# in 30 s)	17.8 ± 4.6	18.0 ± 3.4	17.5 ± 3.9	0.78
UPGO (s)	5.77 ± 1.52	5.21 ± 1.30	5.67 ± 0.87	0.35

Values are presented as means $\pm$ SD

HV functional high-velocity resistance training group, LV low-velocity resistance training group, AC active control group

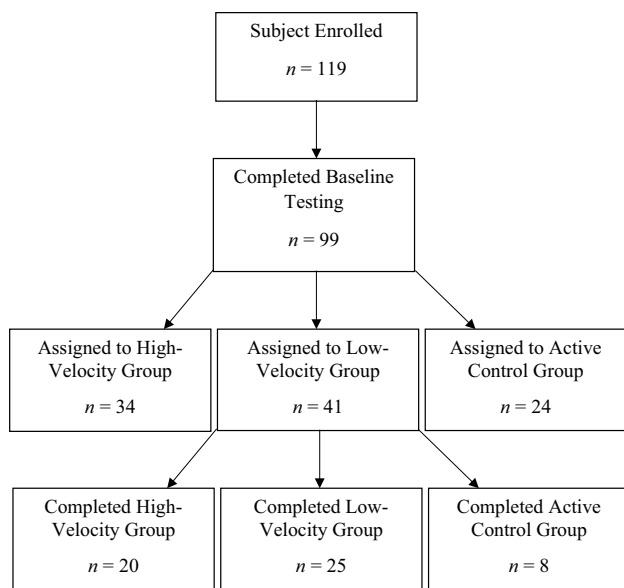


Fig. 1 Flowchart of participant enrollment

Attrition

Attrition was high throughout the study with 53 participants (55%) completing the 48-week intervention and all assessments (Fig. 1). The majority of participants ($n = 35$) dropped out of the study between baseline and 24 weeks, while nine dropped after the 24-week time point. Highest attrition was observed in the AC group (64%) followed by the HV (47%), and LV (31%) groups. The most common reasons for dropping out were lack of interest, health issues, and scheduling conflicts. Eight participants who completed the program were removed from body composition analyses due to inability to complete the DXA scan at 24 weeks.

Body composition

Lean mass was similar between groups at baseline ($F(2, 41) = 0.73$, $p = 0.49$). After the 48-week intervention, a significant group-by-time interaction was observed for lean mass ($F(2, 40) = 3.72$, $p = 0.03$) while covarying out baseline scores. Post hoc analyses did not reveal significant changes in lean-tissue mass within each group over time (Fig. 2). No other significant interaction or time effects were observed for fat mass (Fig. 3) or bone mineral content (Fig. 4).

Physical function

No differences existed for any physical function assessment at baseline ($p > 0.05$). A significant group-by-time

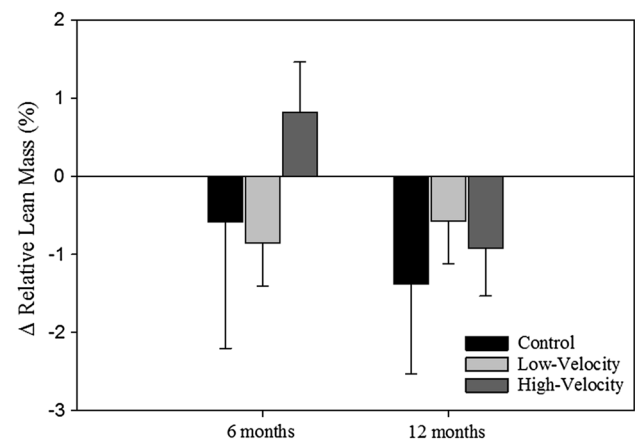


Fig. 2 Changes in lean mass

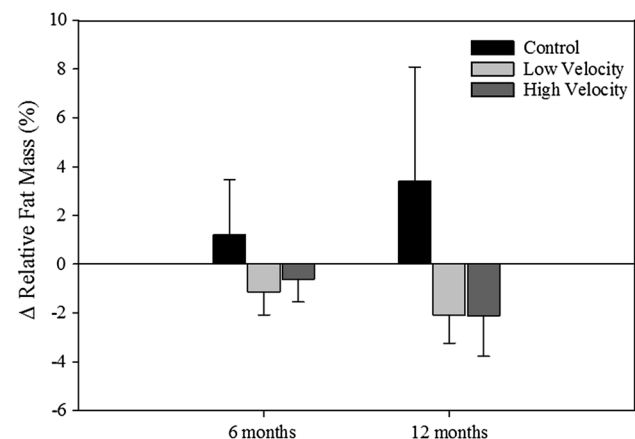


Fig. 3 Changes in fat mass

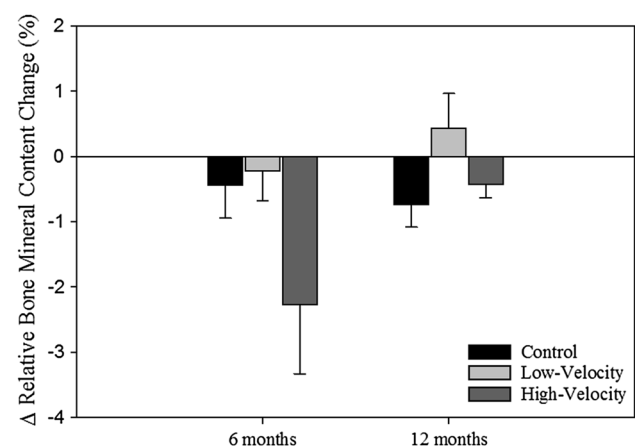


Fig. 4 Changes in bone mineral content

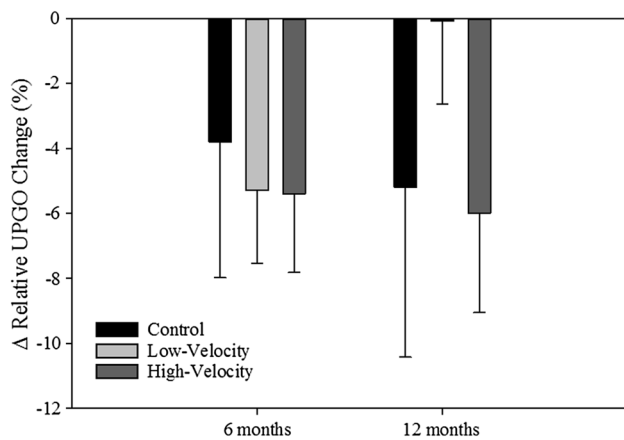


Fig. 5 Changes in eight-foot up-and-go performance

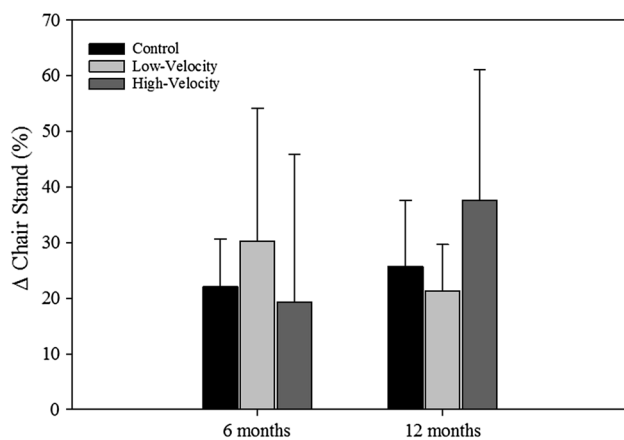


Fig. 6 Changes in chair stand performance

interaction was observed for the eight-foot up-and-go test, ($F(2, 41) = 3.73$, $p = 0.03$). From baseline to 48 weeks, the HV and AC groups improved up-and-go performance by 5 and 6%, respectively (Fig. 5). Although there was no significant group-by-time interaction for chair stand performance, there was a trend for an increase over time for all three groups (Fig. 6). Additionally, arm curl performance did not significantly change in any group over time ($p = 0.90$); however, large improvements were observed in both exercise training groups from baseline to 24 weeks ($p < 0.017$) with no further improvements after this time point.

Strength and power

Significant increases in muscular strength occurred in both of the exercise training groups from baseline to 24 weeks ($p < 0.05$) with no significant changes in the AC group. The greatest increase followed LV training; after 24 weeks of training, this group increased total strength by 43%. HV participants increased total strength by 27% in the first 24 weeks (Table 2). Only the LV group significantly increased from 24 to 48 weeks ($p < 0.001$; Fig. 7).

Muscular power was similar between groups at baseline ($p = 0.66$). No significant differences between any group over time ($F(2, 17) = 1.02$, $p = 0.38$). However, muscular power was reduced in the AC by 10% from baseline to 24 weeks ($p = 0.048$) and remained unchanged in the two training groups. After 48 weeks, muscular power was not different from baseline for any group.

Table 2 Muscular strength changes

Variable	Baseline			Week 24			Week 48		
	HVRT	LVRT	CON	HVRT	LVRT	CON	HVRT	LVRT	CON
Chest press (kg)	3.88 ± 1.70	4.31 ± 1.97	3.86 ± 0.89	4.83 ± 1.46*	4.60 ± 2.37	4.77 ± 1.17	5.12 ± 1.87	4.36 ± 0.62	4.24 ± 0.74
Bent-over row (kg)	10.45 ± 6.11	10.39 ± 4.83	9.33 ± 4.35	11.84 ± 4.83	10.91 ± 5.00	11.01 ± 4.86	12.76 ± 4.84	12.22 ± 4.64	10.00 ± 2.61
Lateral Raise (kg)	3.67 ± 1.64	3.96 ± 1.54	3.60 ± 1.40	4.74 ± 1.30	4.65 ± 2.09	4.38 ± 1.08	5.13 ± 2.34	4.60 ± 1.25	4.24 ± 0.94
Biceps curls (kg)	6.27 ± 2.68	7.90 ± 8.90	5.64 ± 2.31	7.38 ± 2.68	7.13 ± 2.88	6.46 ± 2.02	8.21 ± 4.66	7.06 ± 1.72	6.43 ± 1.30
Triceps ext (kg)	5.17 ± 2.63	4.91 ± 1.77	4.61 ± 1.81	5.96 ± 1.51	6.30 ± 4.77	5.45 ± 0.82	6.45 ± 2.29	5.96 ± 1.45	5.39 ± 0.96
Knee curl (kg)	10.18 ± 5.90	10.39 ± 3.86	8.92 ± 3.68	15.88 ± 6.86*	12.85 ± 5.27	13.54 ± 3.82	17.94 ± 6.59	16.90 ± 5.47**	14.35 ± 4.53
Total Strength (kg)	36.91 ± 18.04	37.17 ± 14.31	31.30 ± 10.42	48.83 ± 18.96*	44.17 ± 18.96	44.60 ± 13.07*	52.85 ± 20.87	50.25 ± 16.76	43.44 ± 11.21

Values are reported as means ± SD

HVRT high-velocity resistance training group, LVRT low-velocity resistance training, AC active control group.

*Significantly different from baseline ($p < 0.015$)

**Significantly different from 24 weeks ($p < 0.015$)

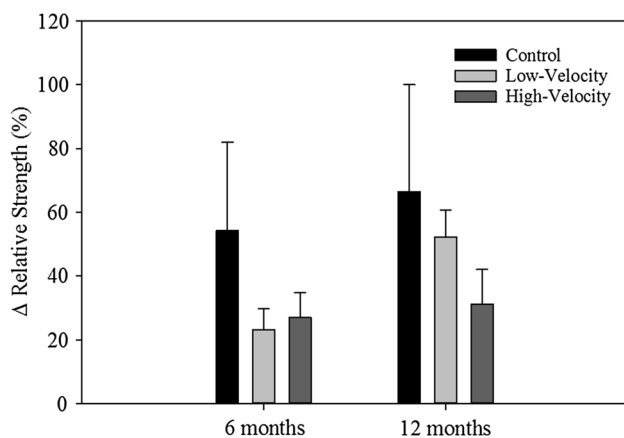


Fig. 7 Changes in muscular

Discussion

Significant group-by-time changes were observed in lean-tissue mass and up-and-go performance after the 48-week intervention. For up-and-go, performance was improved in the AC group; however, decreased in the LV group and remained unchanged in the HV group after 12 months. No other significant changes were observed in any group.

Changes in body composition were modest, but in the expected direction after exercise training [42]. Effects sizes revealed small effects from baseline to 48 weeks in all groups. However, the absolute change was similar to other studies after resistance training among older adults [43, 44].

In the present investigation, the LV group experienced a decline in dynamic balance performance as assessed by the eight-foot up-and-go assessment, while the HV and AC groups remained unchanged. These results could be an outcome of specificity of training. During the LV training program, participants performed all exercises at a rate of 2 s for each phase (concentric and eccentric) for each lift. Other investigators have reported no changes in up-and-go performance after training LV resistance training [16, 45]; however, improvements have been reported after HV [16]. Additionally, initial changes in functional lower-body strength (chair stand test) were large (from baseline to 24 weeks) in all groups (10–16%), but improvements did not extend beyond the initial familiarization period. These results are similar to other studies reporting marginal increases in chair stand performance after HV resistance training when compared to LV resistance training [16].

A major strength of this project was the location of training. All resistance training classes were conducted in the community, in close proximity to each participant. Additionally, the equipment used is easily accessible from any department or sporting goods store. These two aspects of the study eliminated important barriers [46] to exercise participation

for older adults. Furthermore, resistance training interventions have documented benefits for LM improvement [24, 26–28, 47]; however, this is the first intervention to observe changes in body composition in a community-based setting. The location was imperative to translate results of laboratory studies into a community-based intervention. There are notable limitations in the present investigation. The subject sample included only healthy, independent-living older adults; thus, care should be used when generalizing results to other older adult populations. Additionally, the speed of movement was not quantified during each training session. As an alternative, each subject in the HV group (24–48 weeks) was instructed to perform each exercise as quickly as possible as compared to the LV group who performed the exercises at 2-s increments (concentric/eccentric). Attrition in the current study was high (46%). Other studies report attrition ranging from 10 to 18% [15, 16, 32, 33, 48]. The greatest attrition occurred in AC group. Based on anecdotal evidence, although randomly assigned, the AC group expressed a lack of interest in continuing in the study due to being assigned to the AC group. Despite the best attempts from the research team, the participants could not be convinced to remain in the study.

In conclusion, high-velocity resistance training using free weight resistance appears to be a safe and effective alternative to lower velocity training resulting in no injuries throughout the 48-week study duration. High-velocity resistance training produced similar body composition results to low-velocity training [17, 29, 30, 48–52] after the exercise intervention. However, the high-velocity resistance training group had greater increase in 8-foot up-and-go performance—a commonly used measure of physical function. We recommend further investigations to determine the optimal training intensity for free weight high-velocity resistance training and explore the use of training at higher velocities among clinical populations.

Compliance with ethical standards

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

Statement of human and animal rights This study was approved by the Institutional Review Board at a Midwest Regional University.

Informed consent All participants provided written informed consent.

References

1. Fried LP, Ferrucci L, Darer J et al (2004) Untangling the concepts of disability, frailty, and comorbidity: implications for improved targeting and care. *J Gerontol Ser A Biol Sci Med Sci* 59:M255–M263. <https://doi.org/10.1093/gerona/59.3.M255>

2. Runge M, Rittweger J, Russo CR et al (2004) Is muscle power output a key factor in the age-related decline in physical performance? A comparison of muscle cross section, chair-rising test, and jumping power. *Clin Physiol Funct Imaging* 24:335–340
3. Rose DJ, Jones CJ, Lucchese N (2002) Predicting the probability of falls in community-residing older adults using the 8-foot up-and-go: a new measure of functional mobility. *J Aging Phys Activ* 10:466–475
4. Shumway-Cook A, Silver IF, LeMier M et al (2007) Effectiveness of a community-based multifactorial intervention on falls and fall risk factors in community-living older adults: a randomized, controlled trial. *J Gerontol MED SCI* 62A:1420–1427
5. Tinetti ME, Speechley M, Ginter SF (1988) Risk factors for falls among elderly persons living in the community. *N Engl J Med* 319:1701–1707
6. Janssen I, Heymsfield SB, Ross R (2002) Low relative skeletal muscle mass (sarcopenia) in older persons is associated with functional impairment and physical disability. *J Am Geriatr Soc* 50:889–896
7. English KL, Paddon-Jones D (2010) Protecting muscle mass and function in older adults during bed rest. *Curr Opin Clin Nutr Metab Care* 13:34–39. <https://doi.org/10.1097/MCO.0b013e328333aa66>
8. Baumgartner RN, Waters DL, Gallagher D et al (1999) Predictors of skeletal muscle mass in elderly men and women. *Mech Ageing Dev* 107:123–136. [https://doi.org/10.1016/S0047-6374\(98\)00130-4](https://doi.org/10.1016/S0047-6374(98)00130-4)
9. Gianoudis J, Bailey CA, Daly RM (2014) Associations between sedentary behaviour and body composition, muscle function and sarcopenia in community-dwelling older adults. *Osteoporos Int* 26:571–579. <https://doi.org/10.1007/s00198-014-2895-y>
10. Ryu M, Jo J, Lee Y et al (2013) Association of physical activity with sarcopenia and sarcopenic obesity in community-dwelling older adults: the Fourth Korea National Health and Nutrition Examination Survey. *Age Ageing*. <https://doi.org/10.1093/agein/afq063>
11. Centers for Disease Control and Prevention (2013) The state of aging and health in America in 2013. Atlanta, GA
12. King AC (2001) Interventions to promote physical activity by older adults. *J Gerontol Ser A Biol Sci Med Sci* 56:36–46
13. Bennett JA, Winters-Stone K (2011) Motivating older adults to exercise: what works? *Age Ageing* 40:148–149. <https://doi.org/10.1093/ageing/afq182>
14. Sayers SP, Gibson K (2012) Effects of high-speed power training on muscle performance and braking speed in older adults. *J Aging Res*. <https://doi.org/10.1155/2012/426278>
15. Bean JF, Herman S, Kiely DK et al (2004) Increased velocity exercise specific to task (InVEST) training: a pilot study exploring effects on leg power, balance, and mobility in community-dwelling older women. *J Am Geriatr Soc* 52:799–804
16. Bottaro M, Machado SN, Nogueira W et al (2007) Effect of high versus low-velocity resistance training on muscular fitness and functional performance in older men. *Eur J Appl Physiol* 99:257–264. <https://doi.org/10.1007/s00421-006-0343-1>
17. Bean JF, Kiely DK, LaRose S et al (2009) Increased velocity exercise specific to task training versus the National Institute on Aging's strength training program: changes in limb power and mobility. *J Gerontol Biol Sci Med Sci* 64:983–991. <https://doi.org/10.1093/gerona/glp056>
18. Borst SE (2004) Interventions for sarcopenia and muscle weakness in older people. *Age Ageing* 33:548–555. <https://doi.org/10.1093/agein/afh201>
19. Fiatarone MA, Marks EC, Ryan ND et al (1990) High-intensity strength training in nonagenarians. *J Am Med Assoc* 263:3029–3034
20. Fiatarone MA, O'Neill EF, Ryan ND et al (1994) Exercise training and nutritional supplementation for physical frailty in very elderly people. *N Engl J Med* 330:1769–1775. <https://doi.org/10.1056/NEJM19940623302501>
21. Joseph LJ, Davey SL, Evans WJ, Campbell WW (1999) Differential effect of resistance training on the body composition and lipoprotein-lipid profile in older men and women. *Metab Clin Exp* 48:1474–1480
22. Latham NK, Bennett DA, Stretton CM et al (2004) Systematic review of progressive resistance strength training in older adults. *J Gerontol Ser A Biol Sci Med Sci* 59:48–61. <https://doi.org/10.1093/gerona/59.1.M48>
23. Taaffe DR, Pruitt L, Pyka G et al (1996) Comparative effects of high- and low-intensity resistance training on thigh muscle strength, fiber area, and tissue composition in elderly women. *Clin Physiol (Oxford Engl)* 16:381–392
24. Vincent KR, Braith RW, Feldman RA et al (2002) Resistance exercise and physical performance in adults aged 60 to 83. *J Am Geriatr Soc* 50:1100–1107
25. Martyn-St James M, Carroll S (2006) High-intensity resistance training and postmenopausal bone loss: a meta-analysis. *Osteoporos Int* 17:1225–1240. <https://doi.org/10.1007/s00198-006-0083-4>
26. Binder EF, Yarasheski KE, Steger-May K et al (2005) Effects of progressive resistance training on body composition in frail older adults: results of a randomized, controlled trial. *J Gerontol Ser A Biol Sci Med Sci* 60:1425–1431. <https://doi.org/10.1093/gerona/60.11.1425>
27. Kalapotharakos VI, Michalopoulou M, Godolias G et al (2004) The effects of high- and moderate-resistance training on muscle function in the elderly. *J Aging Phys Act* 12:131–143
28. Chalé A, Cloutier GJ, Hau C et al (2012) Efficacy of whey protein supplementation on resistance exercise-induced changes in lean mass, muscle strength, and physical function in mobility-limited older adults. *J Gerontol Ser A Biol Sci Med Sci*. <https://doi.org/10.1093/gerona/gls221>
29. Henwood TR, Riek S, Taaffe DR (2008) Strength versus muscle power-specific resistance training in community-dwelling older adults. *J Gerontol Ser A Biol Sci Med Sci* 63:83–91
30. Reid KF, Callahan DM, Carabello RJ et al (2008) Lower extremity power training in elderly subjects with mobility limitations: a randomized controlled trial. *Aging Clin Exp Res* 20:337–343
31. Kraemer WJ, Adams K, Cafarelli E et al (2002) American College of Sports Medicine position stand. Progression models in resistance training for healthy adults. *Med Sci Sports Exerc* 34:364–380
32. de Vos NJ, Singh NA, Ross DA et al (2005) Optimal load for increasing muscle power during explosive resistance training in older adults. *J Gerontol Ser A Biol Sci Med Sci* 60:638–647
33. Caserotti P, Aagaard P, Buttrup Larsen J et al (2008) Explosive heavy-resistance training in old and very old adults: Changes in rapid muscle force, strength and power. *Scand J Med Sci Sports* 18:773–782. <https://doi.org/10.1111/j.1600-0838.2007.00732.x>
34. Tombaugh TN, McIntyre NJ (1992) The mini-mental state examination: a comprehensive review. *J Am Geriatr Soc* 40:922–935
35. Rikli RE, Jones CJ (2001) Senior fitness test manual. Human Kinetics, Champaign, IL
36. Hesseberg K, Bentzen H, Bergland A (2015) Reliability of the senior fitness test in community-dwelling older people with cognitive impairment. *Physiother Res Int J Res Clin Phys Ther* 20:37–44. <https://doi.org/10.1002/pri.1594>
37. Pescatello LS, Arena R, Riebe D, Thompson PD (eds) (2014) ACSM's guidelines for exercise testing and prescription, 9th edn. Wolters Kluwer/Lippincott, Williams, and Wilkins, Baltimore, MD
38. Roberts HC, Denison HJ, Martin HJ et al (2011) A review of the measurement of grip strength in clinical and epidemiological

- studies: towards a standardised approach. *Age Ageing* 40:423–429. <https://doi.org/10.1093/ageing/afr051>
39. Sallinen J, Stenholm S, Rantanen T et al (2010) Hand-grip strength cut points to screen older persons at risk for mobility limitation. *J Am Geriatr Soc* 58:1721–1726. <https://doi.org/10.1111/j.1532-5415.2010.03035.x>
 40. Giampaoli S, Ferrucci L, Cecchi F et al (1999) Hand-grip strength predicts incident disability in non-disabled older men. *Age Ageing* 28:283–288. <https://doi.org/10.1093/ageing/28.3.283>
 41. Gray M, Paulson S (2014) Developing a measure of muscular power during a functional task for older adults. *BMC Geriatr* 14:145–150
 42. Galvão DA, Taaffe DR (2005) Resistance exercise dosage in older adults: single- versus multiset effects on physical performance and body composition. *J Am Geriatr Soc* 53:2090–2097. <https://doi.org/10.1111/j.1532-5415.2005.00494.x>
 43. Tieland M, Dirks ML, van der Zwaluw N, Verdijk LB, van de Rest O, de Groot L, van Loon LJC (2012) Protein supplementation increases muscle mass gain during prolonged resistance-type exercise training in frail elderly people: a randomized, double-blind, placebo-controlled trial. *J Am Med Dir Assoc* 13:713–719. <https://doi.org/10.1016/j.jamda.2012.05.020>
 44. Wernbom M, Augustsson J, Thomee R (2007) The influence of frequency, intensity, volume and mode of strength training on whole muscle cross-sectional area in humans. *Sports Med* 37:225–264
 45. De Vreede PL, Samson MM, Van Meeteren NLU et al (2005) Functional-task exercise versus resistance strength exercise to improve daily function in older women: a randomized, controlled trial. *J Am Geriatr Soc* 53:2–10. <https://doi.org/10.1111/j.1532-5415.2005.53003.x>
 46. Bethancourt HJ, Rosenberg DE, Beatty T et al (2014) Barriers to and facilitators of physical activity program use among older adults. *Clin Med Res* 12:10–20. <https://doi.org/10.3121/cmr.2013.1171>
 47. Fiatarone MA, Marks EC, Ryan ND et al (1990) High-intensity strength training in nonagenarians. Effects on skeletal muscle. *JAMA J Am Med Assoc* 263:3029–3034
 48. Sayers SP, Bean J, Cuoco A et al (2003) Changes in function and disability after resistance training: does velocity matter?: a pilot study. *Am J Phys Med Rehabil Assoc Acad Physiatr* 82:605–613. <https://doi.org/10.1097/01.phm.0000078225.71442.b6>
 49. Marsh AP, Miller ME, Rejeski WJ et al (2009) Lower extremity muscle function after strength or power training in older adults. *J Aging Phys Act* 17:416–443
 50. Fielding RA, LeBrasseur NK, Cuoco A et al (2002) High-velocity resistance training increases skeletal muscle peak power in older women. *J Am Geriatr Soc* 50:655–662. <https://doi.org/10.1046/j.1532-5415.2002.50159.x>
 51. Miszko TA, Cress ME, Slade JM et al (2003) Effect of strength and power training on physical function in community-dwelling older adults. *J Gerontol Med Sci* 58A:171–175
 52. Sayers SP, Guralnik JM, Thombs LA, Fielding RA (2005) Effect of leg muscle contraction velocity on functional performance in older men and women. *J Am Geriatr Soc* 53:467–471. <https://doi.org/10.1111/j.1532-5415.2005.53166.x>